An effective deep CNN-LSTM based intrusion detection system for network security

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***Abstract*— Deep learning (DL) models have become prevalent and consistently exhibit outstanding performance across diverse domains, particularly in information security. As a subset of machine learning (ML), DL has proven adept at handling complex, unstructured, and large-scale data, making it well-suited to tackle the evolving challenges of cybersecurity. By leveraging DL techniques, we can develop robust intrusion detection systems (IDS) to counter a wide range of network attacks, serving as a crucial defense against hackers and cybercriminals. The integration of DL into IDS holds immense potential for significantly enhancing network security and mitigating cyber risks in various real-world scenarios. This work proposes an IDS architecture that integrates Convolutional Neural Network (CNN) for spatial feature extraction and Long Short-Term Memory (LSTM) for temporal feature extraction and sequential analysis to predict and classify malicious cyberattacks in IoT network traffic. Our experiments with the CSE-CIC-IDS2018 dataset, the latest comprehensive network traffic dataset, demonstrate the superiority of our approach in terms of accuracy, precision, recall, and F1 score. Our model achieves an unprecedented accuracy of over 99% in both binary and multiclass classification, surpassing existing efforts in the literature.**

Keywords*— *Dl, IDS, CNN, LSTM, RNN, Network Traffic

I. Introduction

The extensive adoption of DL has revolutionized many sectors, spanning computer vision, healthcare, natural language processing, speech and visual recognition [1]. DL, a branch of ML, entails training artificial neural networks with vast datasets to discern complex patterns and forecast outcomes. Its capacity to autonomously discern complex patterns in data has driven notable progress in diverse fields [2].

DL has gained significant traction in recent years, particularly in the IoT and cybersecurity domains [3], where it is employed for assignments like intrusion detection, malware analysis, and threat intelligence. With IoT environments characterized by interconnected devices generating vast amounts of data, there are various security challenges to address. The application of DL in IDS improves their effectiveness and robustness in fortifying defenses against emerging threats and safeguarding interconnected systems in an increasingly digital world. DL-based IDS can analyze diverse data streams from IoT devices, including network traffic and sensor data, to detect anomalies and potential security breaches in real-time [4].

Recent advancements in IDS have been driven by the use of DL algorithms. DL, which encompasses architectures such as Recurrent Neural Networks (RNN) like Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), as well as Convolutional Neural Networks (CNN), Deep Belief Networks (DBN), and Deep Neural Networks (DNN) [5], has significantly enhanced the capabilities of IDS. These DL approaches and their combinations have demonstrated remarkable effectiveness in detecting and classifying network intrusions, surpassing the performance of traditional ML models [6].

This work introduces an IDS based on a DL model that combines CNN and LSTM networks, a variant of RNN, to capture spatial and temporal features effectively in the data. The proposed models have been evaluated using the CSE-CIC-IDS2018 dataset, which is the newest dataset of network traffic. Our results demonstrate that integrating these two DL algorithms significantly improves performance, enhancing the accuracy and resilience of the IDS against threats and attacks.

This is how the rest of the document is arranged. A summary of previous literature evaluations is given in Section II. Section III present a brief background about the CNN and LSTM model architecture, along with a detailed discussion of our proposed CNN-LSTM model. The CSE-CIC-IDS2018 dataset which was utilized to train our model, metrics and model setting are evaluated in Section IV. In Section V, the findings of our study are given and discussed. This essay's conclusion, Section VI, addresses potential study avenues.

II. Related Work

Significant research efforts have been dedicated to IDS for IoT networks, focusing on DL methodologies. In this section, we introduce the existing works that have utilized the CSE-CIC-IDS2018 dataset.

Ferrag et al. [7] presents a comprehensive comparative study of seven deep learning models namely DNN, RNN, CNN,

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Restricted Boltzmann Machine (RBM), Deep Belief Network (DBN), Deep Boltzmann Machine (DBM), and Deep Autoencoder (DAE) with the aim of enhancing the cybersecurity of IDS. It provides detailed explanations of each DL approach and describes 35 different datasets used in the study. The performance evaluation covers both binary and multiclass classification tasks using the CSE-CICIDS2018 and the Bot-IoT datasets, demonstrating highly promising results with an accuracy exceeding 98%.

In [8], the authors proposed an optimized custom RC-NN model for intrusion detection, which combines CNN and LSTM with the meta-heuristic Ant Lion optimization algorithm. This method was evaluated in binary classification using the DARPA and CSE-CIC-IDS2018 datasets. The proposed approach attained an enhanced classification accuracy of 94% with a reduced error rate of 0.0011.

According to [9], two popular DL approaches, CNN and LSTM, are discussed. The Autoencoder and principal component analysis (PCA) algorithm are used for feature reduction in the original dataset. The experiment was conducted using the CSE-CIC-IDS2018 dataset. The accuracy achieved was approximately 0.9987 for CNN and 0.9983 for LSTM. These results imply that employing feature reduction methods decreased processing time without compromising the accuracy of the overall detection performance.

Ref [10] compared the effectiveness of three distinct DL models: LSTM, GRU, and RNN in addressing the issue of anomaly detection using the CICIDS2017 and CSE-CICIDS2018 datasets.

Another study leveraging the CSE-CIC-IDS2018 datasets was introduced in [11], where the authors utilized a U-Net and Temporal Convolutional Network (TCN) for network attack classification. The experiment also employed an older dataset, the KDD99. The TCN with LSTM achieved 92% and 97% accuracy on KDD99 and CSE-CIC-IDS2018, respectively, while the U-Net model achieved 93% and 94% accuracy on the same datasets. To address class imbalance in time-series data, they implemented the focal loss function in the TCN with LSTM. Their findings demonstrate that TCN with LSTM and U-Net outperform other network architectures for network traffic classification. They also noted that models trained on the old dataset can overfit and perform well on the testing set, but they are less effective on more complex and current data.

Imrana et al. [12] presented a BiLSTM-based IDS. The model was tested using the NSL-KDD dataset and demonstrated strong performance compared to the LSTM model. However, the recognition rate for minority class attacks U2R and R2L remains suboptimal and needs improvement.

III. Proposed Method

We start by exploring the structure of CNN. Then, we delve into the architecture of LSTM. Finally, we provide a detailed overview of our proposed CNN-LSTM model.

A. Convolutional Neural Network (CNN)

Recently, CNNs have achieved significant success and become essential tools in a wide array of domains such as facial recognition, computer vision, voice recognition, object detection, and image classification [13]. They utilize multiple layers of arrays to process data, resulting in their deep structure. CNNs are often designed to extract spatial correlations of features and accelerate the training process by addressing the problem of parameter explosion through weight sharing. As illustrated in Figure 1, CNNs typically consist of three main components: the convolution layer, the pooling layer, and a fully connected layer. The convolution layer extracts features by applying different kernels, the pooling layer reduces feature dimensions through down sampling, and the fully connected layer connects the extracted features to produce classification results [14].

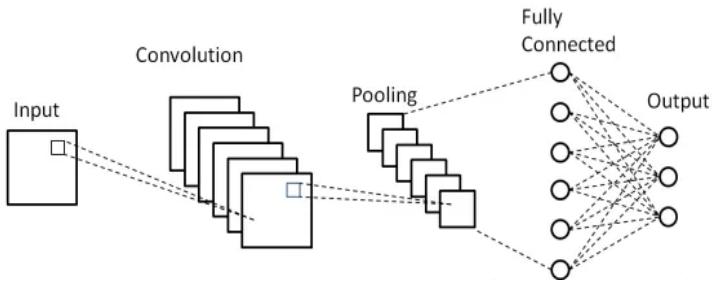

Fig. 1. CNN architecture

B. Long Short-Term Memory (LSTM)

LSTM is an advanced RNN approach developed by Hochreiter et al. [15] to tackle the gradient vanishing problem and address the issue of long-term dependencies in RNNs. LSTM has demonstrated exceptional performance across various applications, including images and video analysis, machine translation, and language modeling [16]. A typical LSTM network architecture consists of a memory cell and three regulators including input gate, output gate, and forget gate that control data flow and feedback within the LSTM. Figure 2 illustrates a simplified architecture of an LSTM cell. The following steps are required to update the activation of a neural unit:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \tag{1}$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \tag{2}$$

$$c_t = f_t * c_{t-1} + i_t * \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \tag{3}$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_{t-1} + b_o) \tag{4}$$

$$h_t = o_t * \tanh(c_t) \tag{5}$$

σ and tanh represent the activation functions, where σ is the sigmoid function and tanh is the hyperbolic tangent function. c denotes the cell state, and i, f, and o are the input, forget, and output gates, respectively. W represents the weight matrix, and b defines the bias vector.

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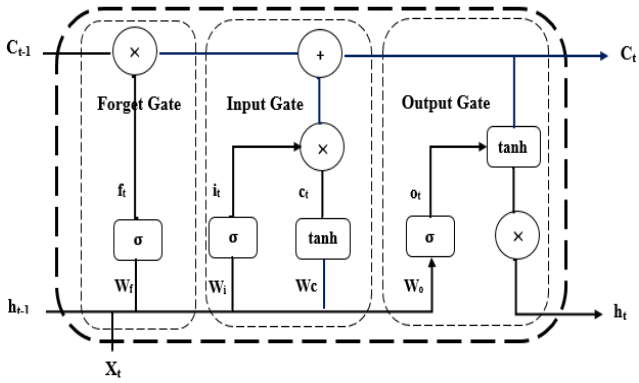

Fig. 2. Structure of LSTM cell

C. Proposed CNN-LSTM

CNN and LSTM are two widely adopted DL methodologies that have demonstrated remarkable performance in intrusion detection for IoT networks [17]. CNN focuses on learning spatial features from network traffic data, whereas LSTM excels in capturing temporal dependencies in time-series data.

Our proposed IDS is based on a DL architecture that combines 1-Dimensional Convolutional Neural Network (1-D CNN) with multiple layers of LSTM. The CNN-LSTM model aims to enhance the detection of various types of cyber-attacks that may target IoT devices.

The model architecture of our network is depicted in Figure 3. It starts with an input layer followed by a 1D CNN with 64 filters, processing an input vector of size 72 × 1 to extract crucial spatial features. Next, a max-pooling layer is applied to reduce data dimensions. Subsequently, a Batch Normalization layer is introduced to normalize parameters between intermediate layers, aiding in faster training. The resulting outputs are then fed into multiple LSTM layers with 64 and 128 units, respectively, to capture temporal dependencies. To prevent overfitting, each LSTM layer is accompanied by a dropout probability of 0.3. Finally, a fully connected layer followed by an output layer is used for classification. The rectified linear unit (ReLU) and softmax activation functions are employed in the model [18].

IV. Experiments Evaluation

A. Dataset

1) Dataset Description: For this study, we utilized one of the most up-to-date publicly available intrusion detection datasets, the CSE-CIC-IDS2018 [19].This is more intricate and represents contemporary real-world network traffic and current attack types. It was developed collaboratively by the Canadian Institute for Cybersecurity (CIC) and the Communications Security Establishment (CSE). The dataset includes seven various assault scenarios and comprises unprocessed PCAP network recordings and CICFlowMeter-V3 processed CSV files, offering 80 features. The network traffic was recorded from Amazon Web Services (AWS) over span of ten days, during various attack scenarios and realistic traffic were evaluated. As a result, the dataset includes instances for both benign network traffic and typical network attacks [20].

2) Dataset Preprocessing: The dataset CSE-CIC-IDS2018 comprises 16 232 942 rows, each cotaining 10 files with 80 features per row. In our experiment, we focused on 3 files containing 3 145 725 data samples and 80 features. Table I provides a detailed description of the dataset used.We excluded 8 features including Timestamp, Protocol, PSH Flag Cnt, Init Fwd Win Byts, Flow Byts/s, Flow Pkts/s, and Label as they did not impact network attack. Dta cleaning involved removing null and infinite values. We allowed infinity values to remain, as they represented highest value in the column with minimal impact on the model evaluation. To avoid these values affecting training, NaN values were replaced with zeros. We also normalized the values to a range of 0 to 1. The labels were encoded using the one-hot encoding technique [21]. A total of 3 135 969 data samples were used for testing and training, with the dataset will split an 80% training set and a 20% testing set. The study's use of both binary and multiclass classification is significant. Therefore, we assigned the Benign label to 0 and the Attack labels to 1 as shown in Table II. For multiclass classification, we grouped the labels into six categories including Benign, Bot, FTP-BruteForce, SSH-Bruteforce, DoS-GoldenEye, and DoS-Slowloris. Tables II and III showed the number of traffic records in our dataset for each type.

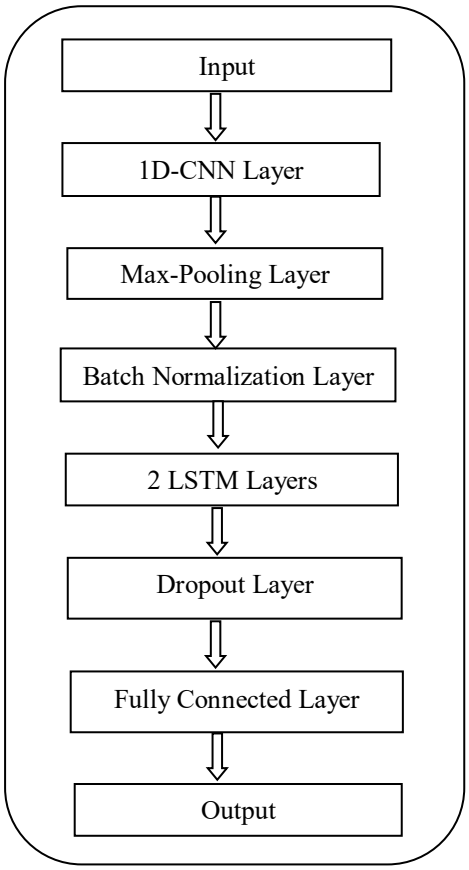

Fig. 3. CNN-LSTM architecture model

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TABLE I. SUMMARY OF THE CSE-CIC-IDS2018 DATASET

Data File	Traffic Type	Number of Samples	Some of features
02-14-2018.csv	Benign	663,808	Dst Port, Protocol, Flow Duration, Tot Fwd Pts, Tot Bwd Pkts, TotLen Fwd PKs, TotLen Bwd Pkts, Fwd Pkt Len Max,
	FTP-BruteForce	193,354	
	SSH-Bruteforce	187,589	
02-15-2018.csv	Benign	988,050	
	DoS-GoldenEye	41,508	
	DoS-Slowloris	1,099	
03-02-2018.csv	Benign	758,334	
	BotAttack	286,191	
Total	3145725		

TABLE II. DETAILS OF CLASS LABELS ON BINARY CLASSIFICATION

Category	Label	Number of samples
Benign	0	2416337
Attack	1	719632

TABLE III. DETAILS OF CLASS LABELS ON MULTI CLASSIFICATION

Category	Label	Number of samples
Benign	0	2426087
Bot	1	286191
FTP-BruteForce	2	193360
SSH-Bruteforce	3	187589
DoS GoldenEye	4	41508
DoS Slowloris	5	10990

B. Metrics

Several metrics are computed to assess IDS performance, considering the following parameters:

- True Positive (TP): Number of correctly classified attack samples.
- False Positive (TN): Number of correctly classified benign samples.
- True Negative (FP): Number of misclassified attack samples.
- False Negative (FN): Number of misclassified benign samples.

The validation of the proposed CNN-LSTM model involved assessing accuracy, recall, precision, and F1 score, as defined in Equations 6, 7, 8, and 9.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{6}$$

$$\Pr ecision = \frac{TP}{TP + FP} \tag{7}$$

$$\operatorname{Re} call = \frac{TP}{TP + FN} \tag{8}$$

$$F1Score = 2 \times \frac{\Pr ecision \times \operatorname{Re} call}{\Pr ecision \times \operatorname{Re} call} \tag{9}$$

C. Model setting

The experiments were carried out in a Windows 10 64-bit operating system environment using Python (version 3.7.13) with TensorFlow (version 2.9.1) and Keras (version 2.9.0) frameworks. Additional library packages such as pandas, numpy, matplotlib, and Sklearn were also utilized. The experiments were executed on a CPU with an Intel(R) Core(TM) i7-8700U CPU clocked at 3.20 GHz, 16 GB of memory, and an Nvidia GeForce GT710 GPU.

In this study, the optimal parameters during the training phase that resulted in the highest accuracy are as follows:

- Number of Epochs: 10
- Batch size: 64
- Learning rate: 0.001
- Optimizer: Adam (Adaptive Moment)
- Cost function: Binary and categorical cross-entropy

V. RESULTS AND DISCUSSIONS

A. Binary Classification

In the binary classification task, our model distinguishes between samples that are attacks and those belonging to a benign class. We assessed the performance of the CNN-LSTM model using a confusion matrix, as shown in Figure 4. Additionally, Figure 5 illustrates the variations in accuracy and loss values of the training and validation sets for binary classification. The binary CNN-LSTM achieved an impressive accuracy of 99.37%. Furthermore, our model demonstrated good precision, recall, and F1 score, with only a few misclassified instances. The evaluation results are summarized in Table IV. Our CNN-LSTM model achieved higher precision and F1-Score for the benign class, approaching 100%. However, these values were slightly lower for attack labels, estimated at 98% and 99%, respectively. Additionally, our model achieved a high recall rate for both benign and attack categories, measured at 99%.

TABLE IV. CNN-LSTM MODELS PERFORMANCE FOR BINARY CLASSIFICATION

Class	Precision	Recall	F1score
Benign	1.00	0.99	1.00
Attack	0.98	0.99	0.99

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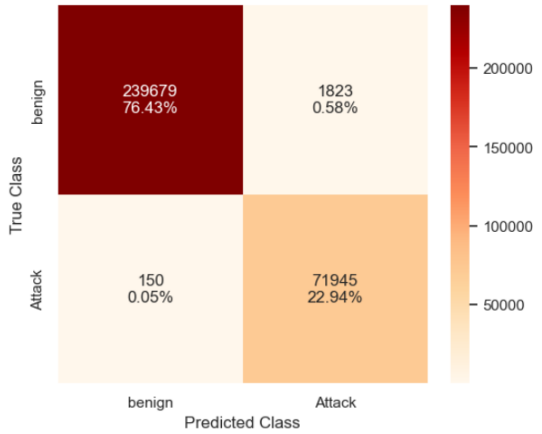
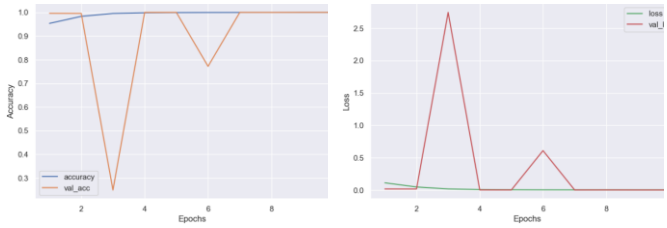
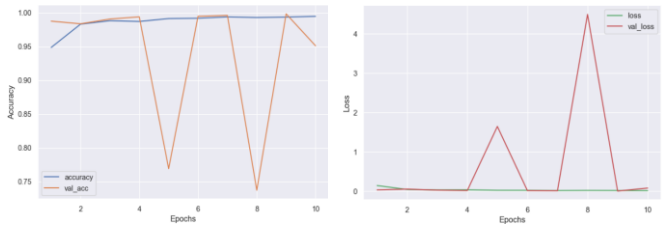


Fig. 4. CNN-LSTM model confusion matrix for binary classification



(a) (b)

Fig. 5. (a) Accuracy. (b) Loss of binary classification



(a) (b)

Fig. 6. (a) Accuracy. (b) Loss of multiclass classification

B. Multiclass Classification

In the multiclass classification, the dataset was divided into benign network traffic or one of the five attacks outlined in Table III. Figure 6 depicts the CNN-LSTM model's performance for multiclass classification in terms of accuracy and loss during training and validation. Figure 7 showcases the confusion matrix used to evaluate our model. The multiclass CNN-LSTM achieved an accuracy of 97.11%. The results of the multiclass evaluation performance are summarized in Table V. Our approach achieved the highest precision and F1-Score for the Benign, FTP-BruteForce, DoS-Slowloris, and Bot, DoS-GoldenEye labels, all reaching 100%. However, the Bot and SSH-Bruteforce classes had the lowest precision and F1-Score, approximately 0.66%, 0.68%, and 0.80%, 0.86%, respectively. Furthermore, the CNN-LSTM successfully detected the Bot, DoS-GoldenEye, and DoS-GoldenEye with a recall value of 100%. Additionally, Benign and FTP-BruteForce were accurately classified, with a detection rate measured at 0.98% and 0.97%, respectively. However, the predictions missed the Bot and SSH-Bruteforce categories.

TABLE V. CNN-LSTM PERFORMANCE FOR MULTICLASS CLASSIFICATION

Class	Precision	Recall	F1score
Benign	1.00	0.98	0.97
Bot	0.66	0.77	0.80
FTP-BruteForce	0.99	0.97	0.98
SSH-Bruteforce	0.68	0.76	0.86
DoS-GoldenEye	1.00	1.00	1.00
DoS-Slowloris	0.99	1.00	1.00

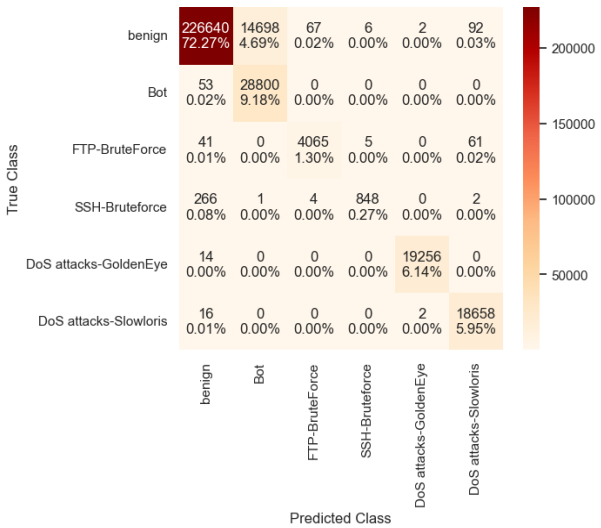


Fig. 7. CNN-LSTM model confusion matrix for multiclass classification

TABLE VI. COMPARISON OF THE EXISTENCE DL APPROACH WITH THE CSE-CIC-IDS2018 DATASET

model	Accuracy	Label Type
HCRNN[22]	97.75%	Binary
DNN[23]	98.97%	
Our CNN-LSTM	**99.37%**	
LSTM[24]	96.20%	Multiclass
CNN[25]	95%	
Our CNN-LSTM	**97.11%**	

C. Comparaison with state of art

Building on our earlier discussion, our CNN-LSTM model underwent rigorous testing with the CSE-CICIDS2018 dataset, evaluating its performance in both binary and multiclass classification. Our model exhibited superior performance in binary classification, achieving an accuracy surpassing 99%. However, in the realm of multiclass classification, our model achieved a still commendable accuracy of 97.11%. Furthermore,

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Table VI provides compelling evidence that our CNN-LSTM model outperformed previously explored DL approaches for IDS on the same dataset.

VI. Conclusion And Future Work

This work endeavors to develop a potent IDS based DL for detecting intruders in IoT networks. Our proposed IDS combines two renowned DL approaches that have recently demonstrated promising results in IoT network intrusion detection. CNN is employed to discern spatial features and automatically identify significant ones from input network traffic data, while LSTM is utilized to extract temporal dependencies between time-series data and act as the classifier. The proposed CNN-LSTM model is assessed using the CSE-CICIDS2018 dataset, a comprehensive recent network traffic dataset, for both binary and multiclass classification. Following the evaluation, the binary classification model outperforms the multiclass classification model in terms of accuracy, recall, precision, and F1 score. Our proposed approach surpasses earlier DL implementations for IDS. In the future, our goal is to assess the proposed architecture using other publicly available IDS datasets, alongside CSE-CIC-IDS2018. We anticipate further research on alternative DL methods for IDS in IoT networks, utilizing various optimization strategies to enhance the detection capabilities of these models for large datasets. We also intend to enhance prediction accuracy by conducting experiments on real IoT applications.

References

[1] Dong, S., Wang, P., & Abbas, K. (2021). A survey on deep learning and its applications. Computer Science Review, 40, 100379.

[2] Shamshirband, S., Fathi, M., Dehzangi, A., Chronopoulos, A. T., & Alinejad-Rokny, H. (2021). A review on deep learning approaches in healthcare systems: Taxonomies, challenges, and open issues. Journal of Biomedical Informatics, 113, 103627.

[3] Li, Y., Zuo, Y., Song, H., & Lv, Z. (2021). Deep learning in security of internet of things. IEEE Internet of Things Journal, 9(22), 22133-22146.

[4] Thakkar, A., & Lohiya, R. (2021). A review on machine learning and deep learning perspectives of IDS for IoT: recent updates, security issues, and challenges. Archives of Computational Methods in Engineering, 28, 3211-3243.

[5] Sarker, I. H. (2021). Deep learning: a comprehensive overview on techniques, taxonomy, applications and research directions. SN Computer Science, 2(6), 420.

[6] Sarker, I. H. (2021). Machine learning: Algorithms, real-world applications and research directions. SN computer science, 2(3), 160.

[7] Ferrag, M.A.; Maglaras, L.; Moschoyiannis, S.; Janicke, H. Deep Learning for Cyber Security Intrusion Detection: Approaches Datasets and Comparative Study. J. Inf. Secur. Appl. 2020, 50, 102419.

[8] T. Thilagam and R. Aruna, Intrusion detection for network based cloud computing by custom RC-NN and optimization, ICT Express, vol. 7, pp. 512-520, 2021.

[9] Antunes, M., Oliveira, L., Seguro, A., Veríssimo, J., Salgado, R., & Murteira, T. (2022, March). Benchmarking deep learning methods for behaviour-based network intrusion detection. In *Informatics* (Vol. 9, No. 1, p. 29). MDPI.

[10] Hai, Tran Hoang, and Le Hai Nam. A practical comparison of deep learning methods for network intrusion detection. 2021 International Conference on Electrical, Communication, and Computer Engineering (ICECCE). IEEE, 2021.

[11] Mezina, A.; Burget, R.; Travieso-González, C.M. Network Anomaly Detection with Temporal Convolutional Network and U-Net model. IEEE Access 2021, 9, 143608–143622.

[12] Y. Imrana, Y. Xiang, L. Ali, and Z. Abdul-Rauf, A bidirectional LSTM deep learning approach for intrusion detection, Expert Systems with Applications, vol. 185, pp. 1–12, 2021.

[13] Li, Z., Liu, F., Yang, W., Peng, S., & Zhou, J. (2021). A survey of convolutional neural networks: analysis, applications, and prospects. IEEE transactions on neural networks and learning systems.

[14] Krichen, M. (2023). Convolutional neural networks: A survey. Computers, 12(8), 151.

[15] Hochreiter, Sepp, and Jürgen Schmidhuber. Long short-term memory. Neural computation 9.8 (1997): 1735-1780.

[16] Lindemann, B., Müller, T., Vietz, H., Jazdi, N., & Weyrich, M. (2021). A survey on long short-term memory networks for time series prediction. Procedia CIRP, 99, 650-655.

[17] Sun, H., Chen, M., Weng, J., Liu, Z., & Geng, G. (2021). Anomaly detection for in-vehicle network using CNN-LSTM with attention mechanism. IEEE Transactions on Vehicular Technology, 70(10), 10880-10893.

[18] Dubey, S. R., Singh, S. K., & Chaudhuri, B. B. (2022). Activation functions in deep learning: A comprehensive survey and benchmark. Neurocomputing.

[19] IDS 2018 Datasets Research Canadian Institute for Cybersecurity UNB. https://www.unb.ca/cic/datasets/ids-2018.html

[20] Leevy, J. L., & Khoshgoftaar, T. M. (2020). A survey and analysis of intrusion detection models based on cse-cic-ids2018 big data. Journal of Big Data, 7(1), 1-19.

[21] Fan, C., Chen, M., Wang, X., Wang, J., & Huang, B. (2021). A review on data preprocessing techniques toward efficient and reliable knowledge discovery from building operational data. Frontiers in Energy Research, 9, 652801.

[22] Khan, M. A. (2021). HCRNNIDS: Hybrid convolutional recurrent neural network-based network intrusion detection system. Processes, 9(5), 834.

[23] Alzughaibi, S., & El Khediri, S. (2023). A Cloud Intrusion Detection Systems Based on DNN Using Backpropagation and PSO on the CSE-CIC-IDS2018 Dataset. Applied Sciences, 13(4), 2276.

[24] Lin, P., Ye, K., & Xu, C. Z. (2019). Dynamic network anomaly detection system by using deep learning techniques. In Cloud Computing–CLOUD 2019: 12th International Conference, Held as Part of the Services Conference Federation, SCF 2019, San Diego, CA, USA, June 25–30, 2019, Proceedings 12 (pp. 161-176). Springer International Publishing.

[25] Pham, V., Seo, E., & Chung, T. M. (2020). Lightweight Convolutional Neural Network Based Intrusion Detection System. J. Commun., 15(11), 808-817.

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